KIDS FULL WORKING MANUSCRIPT
NOT CHILD-VALIDATED
NOT PUBLICATION-READY

KIDS-ELEM1 — Full working manuscript

Age guide: Kindergarten–Grade 2

Review prototype · not child-validated · not publication-ready

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S01 — OBSERVE

figures/FIG-ELEM1-ME-TECH-02.svg

Mira and Bolt sit with a classroom tablet. Look past the glass. Who is using it? What parts touch the world — screen, speakers, charger? What rules decide what is allowed? The bright screen is a window into a bigger system. Draw or write three labels: People · Parts · Rules.

S02 — PREDICT

diagram

Mira taps a lamp icon that only changes this tablet’s brightness. Then she taps Refresh on a class page that sometimes waits. Predict: which tap finishes on this device alone? Which tap might need a message to leave the room? Mark each prediction with a star, not a grade.

S03 — TEST

figures/FIG-ELEM1-ME-TECH-03.svg

Test time with Ping watching the paths. Watch the lamp tap: does the change arrive right away on this device? Watch Refresh: does anything wait? Classify each tap as local or network-ish. If you are unsure, draw a question mark — that is honest science.

S04 — MEASURE

diagram

Measure with simple marks, not fancy clocks. Clap once when the lamp looks brighter. Clap again when Refresh finishes showing new words (if it does). Count the claps between start and finish for each tap. Which job needed more waiting? Waiting is a clue that a message may be traveling.

S05 — EXPLAIN

figures/FIG-ELEM1-ME-TECH-01.svg

Explain the system, not only the screen. Hardware parts sense the touch. Software rules decide what happens next. People choose when to tap and what is OK to share. A local tap can finish on this device. A network tap may send a message out and wait for an answer. Instant highlight does not always mean “finished everywhere.”

S06 — BUILD

diagram

Build a paper system map. Draw the tablet as one box. Add arrows to people, parts, and rules. Add a short loop for a local tap and a longer path for a network tap. Keep this page for your Evidence Folder.

S07 — SECURE

diagram

Shield’s checklist for this unit: use a school-safe or caregiver-approved device; do not install unknown apps; ask before sharing your name, photo, or school; stop if anyone feels rushed. Systems include safety rules on purpose.

S08 — REFLECT

diagram

Misconception check: “The screen is the computer.” Evidence from your map shows people, parts, and rules around the window. Another slogan to rethink: “Every tap goes to the internet.” Your local lamp tap is evidence against that slogan. Kind discussion — no shame.

S09 — TEACH

diagram

Teach a partner or caregiver: technology is a system, and the screen is one part. Use Mira to notice, Bolt for parts, Step for rules, Ping for optional messages, Shield for pause-and-check. Stop anytime. Practice, not ranking.

S10 — OBSERVE

figures/FIG-ELEM1-INSIDE-02.svg

Bolt shows a teaching cutaway — a picture of insides, not a real teardown. Watch a device go busy, then ready. What clues tell you work is happening? Lights, spinners, sounds, or heat a grown-up notices? Observe first. Do not open cases.

S11 — PREDICT

diagram

Predict which inside role handles each job: (1) following steps right now, (2) holding notes for this job, (3) keeping a drawing for tomorrow. Match to CPU, memory, or storage before you see the diagram answers.

S12 — TEST

figures/FIG-ELEM1-INSIDE-01.svg

Test your matches with the role diagram. CPU does steps now. Memory is working notes for the job in progress. Storage is the keep-box for later. Act out with cards: Mira hands Step a “do this next” card (CPU), Bolt holds sticky notes (memory), Ping files a paper in a folder (storage).

S13 — MEASURE

diagram

Measure a tiny difference: open a small drawing you already saved (storage→working), then make one new scribble (CPU + memory), then save again (back to storage). Tally how many adult-approved steps each path needed. Numbers here are counts of steps, not “better scores.”

S14 — EXPLAIN

diagram

Explain without magic. The CPU is not a tiny person. It is a part that follows instructions, one after another — and sometimes more than one job can be in progress on a bigger machine. Memory is for now. Storage is for later. If power stops, working notes can vanish while keep-box files may still be there — adults can help you check safely.

S15 — BUILD

diagram

Build a three-pocket paper sorter labeled CPU · Memory · Storage. Place job stickies into the right pocket. Photograph the paper only if a caregiver says yes — no child accounts.

S16 — SECURE

diagram

Shield reminder: insides are for pictures and models in this book. Real device opening can be unsafe. Ask an adult. Do not share device passcodes while you explore settings menus together.

S17 — REFLECT

diagram

Misconception check: “The CPU is a person inside.” Evidence: it follows instruction cards; it does not feel or know like a classmate. Another rethink: “More always means better.” More parts help some jobs and not others — your tallies show different jobs need different steps.

S18 — TEACH

diagram

Teach someone the three roles with your pockets. Ask them to place one new job card.
Celebrate clear thinking, not speed.

S19 — OBSERVE

figures/FIG-ELEM1-INSTRUCTIONS-02.svg

Press a volume button if an adult says it is OK. What part did you touch? That sensing is hardware. What changed on screen or in sound? Instructions decided the change — that is software. Observe both sides of the same moment.

S20 — PREDICT

diagram

Two apps want attention. Predict what a fair manager program might do: run both at once somehow, or take turns? Step says order matters. Write your prediction before the diagram.

S21 — TEST

figures/FIG-ELEM1-INSTRUCTIONS-01.svg

Test with a human OS game. Two classmates are “apps.” One adult or child is the OS with a turn timer (clap rhythm). Each app gets a short turn to stack blocks. Hardware is the blocks and hands. Software is the rule card: “stack one, then stop.” Watch turn-taking work.

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S22 — MEASURE

diagram

Measure fairness with counts: how many turns did App A get? App B? Did anyone wait with nothing to do? Waiting can be normal when a machine shares time. Record the counts on paper.

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S23 — EXPLAIN

diagram

Hardware notices the world. Software is the recipe — an algorithm is a clear ordered plan. An operating system (OS) schedules which program gets the CPU's attention next. Closing one window may not stop every background job. If something keeps working, ask a caregiver to help check — observe first, then ask.

S24 — BUILD

diagram

Build a four-step algorithm card for a local lamp tap: 1) sense press 2) check which control 3) change brightness 4) show new light. Trade cards and see if a partner can follow your order without guessing. Input → process → output still sits underneath the steps.

S25 — SECURE

diagram

Shield: do not install unknown software. Ask before changing permission switches. Fair access means classmates who need larger text, captions, or different input tools get those tools without being teased.

S26 — REFLECT

diagram

Misconception check: “Closing an app always stops all background work.” Evidence from adult-guided observation may show otherwise — mark what you saw and what you only guessed. Recipes that skip steps cause mix-ups; that is humor, not failure.

S27 — TEACH

diagram

Teach the split: Bolt for hardware, Step for software and turn tickets. Have a partner follow your four-step algorithm exactly. If it breaks, fix the steps together.

S28 — OBSERVE

figures/FIG-ELEM1-MESSAGES-02.svg

Ping draws a path from here to there. Watch a caregiver-approved message or class page load. What stays on this device? What seems to travel? Observe delays without blaming anyone. Networks are shared roads.

S29 — PREDICT

diagram

Predict what happens if a long note is torn into three labeled pieces before it travels. Will the pieces need addresses? Will they always arrive in order? Write yes/no/not sure for each question.

S30 — TEST

figures/FIG-ELEM1-MESSAGES-01.svg

Test with envelope packets. Write one sentence on a strip. Cut into three pieces. Number them 1–2–3 and add “to: Device B.” Hand them along a desk path with two “router” classmates who only forward. Reassemble at the end. Did every piece arrive? What if one waits?

S31 — MEASURE

diagram

Measure reliability with tallies: pieces sent, pieces arrived, pieces late. Reliability is about what actually arrived — not who is “smart.” If a piece is missing, try again; that is normal network thinking.

S32 — EXPLAIN

diagram

A network connects devices so messages can travel. Packets are pieces of a message with addressing information. Paths can use wires or wireless roads like Wi-Fi — Wi-Fi is one kind of road, not the whole internet. The internet is a huge web of networks. Messages do not always arrive; people design retries and checks.

S33 — BUILD

diagram

Build a paper network: two device boxes, one router box, three packet tickets. Tape a path. Keep it for the Evidence Folder.

S34 — SECURE

diagram

Shield at the gate: no unsupervised messaging; ask a trusted adult before sending; never share home address, phone, or passwords; stop if a message feels wrong. Networks are powerful because they connect — and that is why consent matters.

S35 — REFLECT

diagram

Misconception check: “The internet is one wire in the wall.” Evidence: your model needed devices, paths, and packets. Another rethink: “Wi-Fi is the internet.” Wi-Fi can be a road onto larger networks, but the words are not twins. “Messages always arrive” — your late/missing tallies say otherwise.

S36 — TEACH

diagram

Teach a caregiver the packet relay. Let them be a router once. Celebrate clear handoffs.

S37 — OBSERVE

figures/FIG-ELEM1-DATA-02.svg

Mira lays out example cards: three mornings that looked cloudy and felt cool. Observe the pattern without jumping to a big story. What repeats? What is different? Data starts as careful noticing.

S38 — PREDICT

diagram

Predict the next card in a simple pattern (for example: sun, cloud, sun, cloud, ?). Write the guess before flipping the answer card. A prediction is a guess guided by examples — not a proof.

S39 — TEST

figures/FIG-ELEM1-DATA-01.svg

Test the guess. Flip the next card. Were you right? If wrong, add the new example to your data and try again. Machines that “learn” from data also guess from patterns. They can invent wrong answers. People check.

S40 — MEASURE

diagram

Measure with a fair scoreboard: predictions tried, correct, incorrect. No ranking of kids — only counting guesses. If you like, compare two pattern rules and see which fit the examples better.

S41 — EXPLAIN

diagram

Data is information we choose to collect and keep. Patterns help predictions. A model is a tool that uses examples to guess. Generative tools can invent text or pictures that look real but may be false. AI is not a person and does not “know everything.” Checking with a trusted person is part of the job.

S42 — BUILD

diagram

Build a mini data poster: Examples · My prediction · What I checked · What changed. Keep it offline in the Evidence Folder. No child AI chat accounts for this unit.

S43 — SECURE

diagram

Shield: collect only classroom-safe examples (weather icons, block colors) — not private photos of classmates without permission. Do not paste personal stories into public tools. Caregiver mediation required for any digital demo.

S44 — REFLECT

diagram

Misconception check: “AI knows everything” and “AI is a person.” Evidence: your wrong predictions still looked confident until you checked. Confidence is not the same as truth. Another rethink: “Bigger number always means better.” Your tallies measure guesses, not worth.

S45 — TEACH

diagram

Teach a partner your poster. Ask them to make one new prediction and check it. Celebrate careful checking.

S46 — OBSERVE

figures/FIG-ELEM1-SAFE-02.svg

Shield holds a pause hand. Observe a lock screen (do not share real passwords). What is the lock for? Who should be able to open it? Notice fair tools in your room: larger print, captions, quiet corner, different keyboards. Belonging is ordinary.

S47 — PREDICT

diagram

Predict which shares need a yes from a trusted adult: a drawing of a robot; your full name on a public wall; a photo of your face; your school name in a game chat. Mark ask / OK with adult / do not share.

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S48 — TEST

figures/FIG-ELEM1-SAFE-01.svg

Test with role-play cards: Locks · Permissions · Fair access. Mira wants to post a class photo. Shield asks who can see it. Step writes a permission checklist. Bolt adds a large-print copy so more friends can read the rule. Practice calm choices — no scare stories.

S49 — MEASURE

diagram

Measure your habit with a weekly checklist count: times you asked before sharing; times you used a lock with help; times you helped a classmate get a tool they needed. Counts are practice mirrors, not grades.

S50 — EXPLAIN

diagram

Locks and passwords control who may use a device or account. Permissions are rules about what an app or person may see or do. Privacy is choosing who can see information — not hiding forever from everyone who cares for you. Fair access means designing so more people can join. Accessibility is part of fairness, not an optional sticker.

S51 — BUILD

diagram

Build a pocket “Ask first” card with three questions: Who will see this? Do I need an adult yes? Is there a fair way for everyone to join? Decorate without writing real secrets.

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S52 — SECURE

diagram

Shield’s secure practice: protect passwords; do not trade them; refuse unknown installs; no chatting with strangers; stop and tell a trusted adult if something feels wrong. Safety without fear — we practice.

S53 — REFLECT

diagram

Misconception check: “Privacy means hiding forever.” Evidence: you still share with caregivers and teachers when it is safe and needed. “Accessibility is optional extras.” Evidence: fair tools helped someone join your scene. Security is not “hacking practice” — it is protection and consent.

S54 — TEACH

diagram

Teach a caregiver your Ask-first card. Invite them to add one family rule. Easy stop if the talk feels heavy.

S55 — OBSERVE

figures/FIG-ELEM1-BUILD-02.svg

Gather your pages from earlier units: system map, role sorter, algorithm card, packet path, data poster, Ask-first card. Observe what proof you already have. Mira is proud of process, not of beating anyone.

S56 — PREDICT

diagram

Predict which page best shows observation vs guess. Which page best shows a test? Which page shows a safety choice? Write three predictions before sorting.

S57 — TEST

figures/FIG-ELEM1-BUILD-01.svg

Test your folder layout. Use three columns on a cover sheet: What I saw · What I infer (?) · What I still don’t know. Place sticky notes from one favorite investigation into the columns. Inferences keep question marks until tested.

S58 — MEASURE

Measure folder health with counts: number of artifacts; number of question marks still open; number of teach-backs done. Growth means clearer evidence, not higher rank. Unknowns count as honest science.

S59 — EXPLAIN

diagram

Portfolios show how you learned: observing, predicting, testing, measuring, explaining, building, securing, reflecting, and teaching. They are not contests. They are not grades as identity. Careers later use portfolios to show proof — today we practice the habit with kid-safe evidence.

S60 — BUILD

diagram

Build or rebuild your Evidence Folder: cover sheet, six unit pages, one teach-back slip.
Optional: caregiver photos of paper only. No child cloud account required.

S61 — SECURE

Shield: keep private family details out of public displays; ask before photographing faces; store folders in a classroom-safe place. Sharing learning is good when consent is clear.

S62 — REFLECT

diagram

Misconception check: “Portfolio is a contest.” Evidence: your unknowns and question marks are welcome. Compare early guesses to later explanations — improvement is normal. Celebrate repair.

S63 — TEACH

diagram

Teach a short gallery walk: each learner teaches one page. Listeners ask one curious question. Stop anytime. This is practice for explaining systems — the heart of the whole book.